Normal Distribution

Define Log Likelihood Function

Need: - μ (mean) - `sd` (standard deviation)

$\theta = (\mu, \sigma)$

```
ll.norm <- function(theta, x) {
  n <- length(x)
  mu <- theta[1]
  sd <- theta[2]
  print(-n / 2 * log(2 * pi * sd^2) - 1 / (2 * sd^2) * sum((x - mu)^2))
}
```

Generate Data

```
set.seed(69420)
data <- rnorm(1e4, mean = 3, sd = 2)
head(data)
```

```
## [1]  3.5819068  2.7585421 -0.4647484  5.0654662  1.3894668  4.2175822
```

Optimize using optim() to solve for MLE of each param

```
optim(
  par = c(0, 1),
  fn = ll.norm,
  method = "L-BFGS-B",
  lower = c(-Inf, 1e-9),
  control = list(fnscale = -1),
  x = data
)
```

```
## [1] -74603.23
## [1] -74573.18
## [1] -74633.29
## [1] -74482.6
## [1] -74724.25
## [1] -31026.34
## [1] -31019.23
## [1] -31033.46
## [1] -31016.15
## [1] -31036.56
## [1] -29744.52
## [1] -29738.18
## [1] -29750.87
## [1] -29736.46
## [1] -29752.6
## [1] -25736.78
## [1] -25733.02
## [1] -25740.54
## [1] -25734.64
## [1] -25738.92
## [1] -23969
## [1] -23966.52
```

```
## [1] -23971.48
## [1] -23968.96
## [1] -23969.03
## [1] -22494.54
## [1] -22493.35
## [1] -22495.74
## [1] -22495.83
## [1] -22493.26
## [1] -21687.29
## [1] -21687.25
## [1] -21687.33
## [1] -21688.79
## [1] -21685.79
## [1] -21361.73
## [1] -21362.3
## [1] -21361.15
## [1] -21362.48
## [1] -21360.97
## [1] -21199.07
## [1] -21199.33
## [1] -21198.82
## [1] -21198.87
## [1] -21199.29
## [1] -21186.55
## [1] -21186.43
## [1] -21186.68
## [1] -21186.64
## [1] -21186.47
## [1] -21182.55
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## $par
## [1] 3.005513 2.012373
##
## $value
## [1] -21182.53
```

```
##
## $counts
## function gradient
##       14       14
##
## $convergence
## [1] 0
##
## $message
## [1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```

Weibull Distribution

Define Log Likelihood Function

```r
ll.weibull <- function(theta, x) {
  n <- length(x)
  lambda <- theta[1]
  k <- theta[2]
  n * log(k) - n * k * log(lambda) + (k - 1) * sum(log(x)) - sum((x / lambda)^k)
}
```

Generate Data

```r
set.seed(69420)
data <- rweibull(1e6, scale = 69, shape = 420)
head(data)
```

```
## [1] 68.88190 69.12703 68.96215 69.04138 69.19030 69.00712
```

Optimize using optim() to solve for MLE of each param

```r
optim(
  par = c(1, 0),
  fn = ll.weibull,
  method = "L-BFGS-B",
  lower = c(1e-9, 1e-9),
  control = list(fnscale = -1),
  x = data
)
```

```
## $par
## [1]  69.00009 420.58424
##
## $value
## [1] 231845.8
##
## $counts
## function gradient
##       50       50
##
## $convergence
## [1] 0
##
```

```
## $message
## [1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```